Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

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1 Introduction

2 Institutional Background

2.1 Ohio Municipal Income Tax

2.2 Pennsylvania Earned Income Tax

3 Data

3.1 LODES Origin-Destination Data

3.2 Municipal Tax Rate Panels

3.3 Municipal Financial Reports

3.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Ohio (2002–2019)					
share_live_in					
Municipal income tax rate (%)					
Total jobs					
Resident workers					
Panel B: Pennsylvania (2002–2019)					
share_live_in					
Resident EIT rate (%)					
Total jobs					
Resident workers					
Panel dimensions					
OH municipality × years					
PA municipality × years					

Notes:

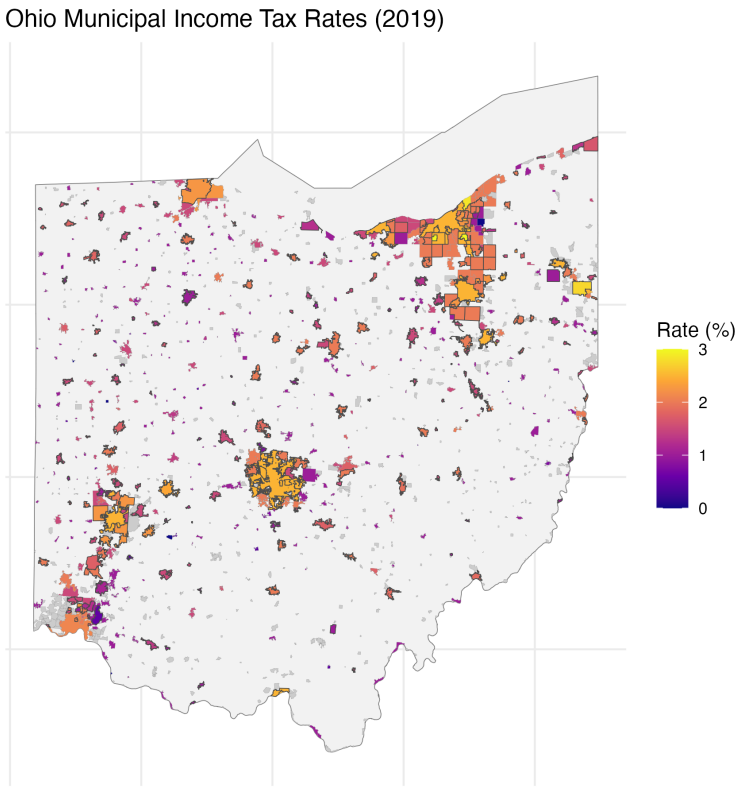


Figure 1: Ohio Municipal Income Tax Rates (2019)

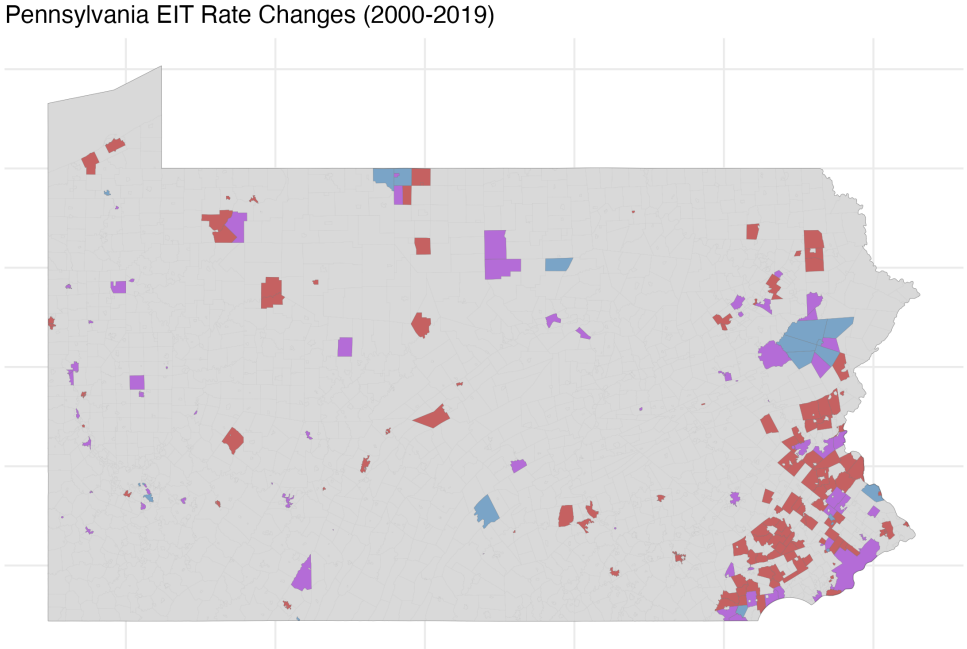


Figure 2: Pennsylvania EIT Rate Changes (2000–2019)

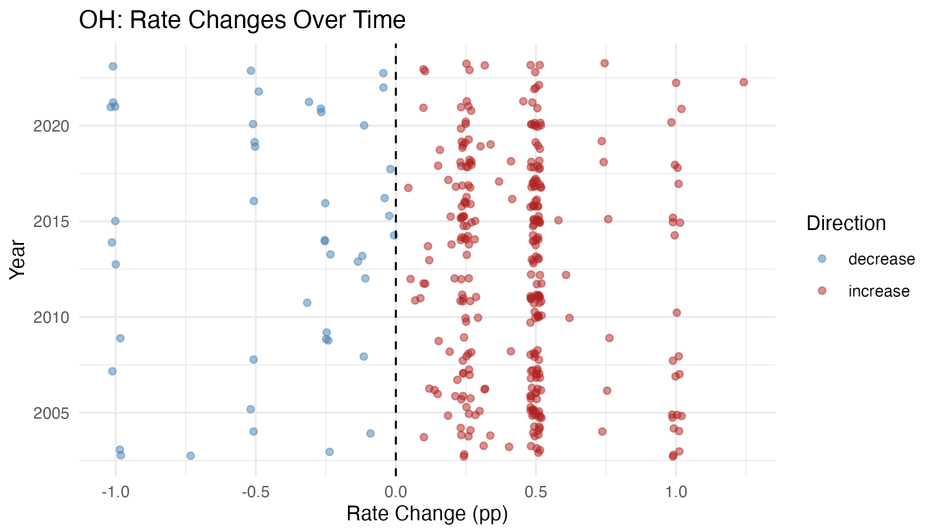


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

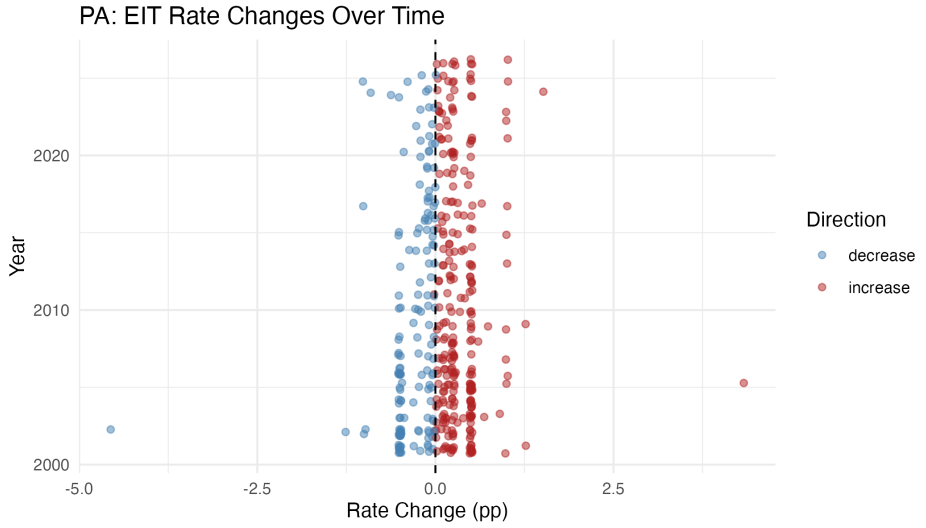


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

4 Empirical Strategy

4.1 Baseline Specification

$$\text{share_live_in}_{it} = \alpha_i + \gamma_t + \beta \cdot \text{rate}_{it} + \varepsilon_{it} \tag{1}$$

where:

$\text{share_live_in}_{it}$ = fraction of workers in municipality i at time t who also reside there

α_i, γ_t = municipality and year fixed effects

rate_{it} = municipal income tax rate

4.2 Dose-Response Identification

$$\text{share_live_in}_{it} = \alpha_i + \gamma_t + \delta \cdot \text{post}_{it} + \beta \cdot (\text{rate}_{it} \times \text{post}_{it}) + \varepsilon_{it} \tag{2}$$

where:

post_{it} = indicator for post-first-increase period

δ captures common post-treatment effects (fiscal stress)

β captures the tax-specific channel: does rate magnitude matter beyond binary treatment?

4.3 Sun-Abraham Event Study

4.4 Stacked Difference-in-Differences

5 Results

5.1 Main Results

Table 2: Main Results: Effect of EIT Rate on share_live_in

	(1)	(2)	(3)	(4)
	PA TWFE	PA Dose	PA Post + Dose	OH Stacked
Coefficients				
rate / resident_eit				
rate × post				
post				
Diagnostics				
N				
R^2				
Within R^2				
Dep. var. mean				
Fixed Effects				
Clusters				
Sample	All	Treated	Treated	Border

Notes: Standard errors clustered at the municipality level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

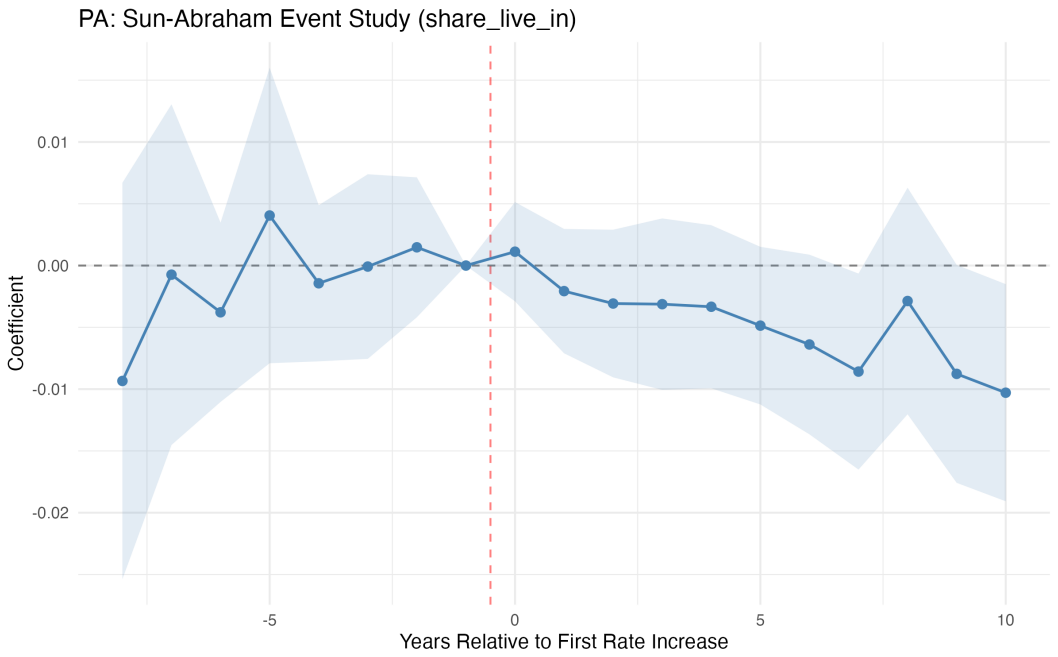


Figure 5: Pennsylvania: Sun-Abraham Event Study (share_live_in)

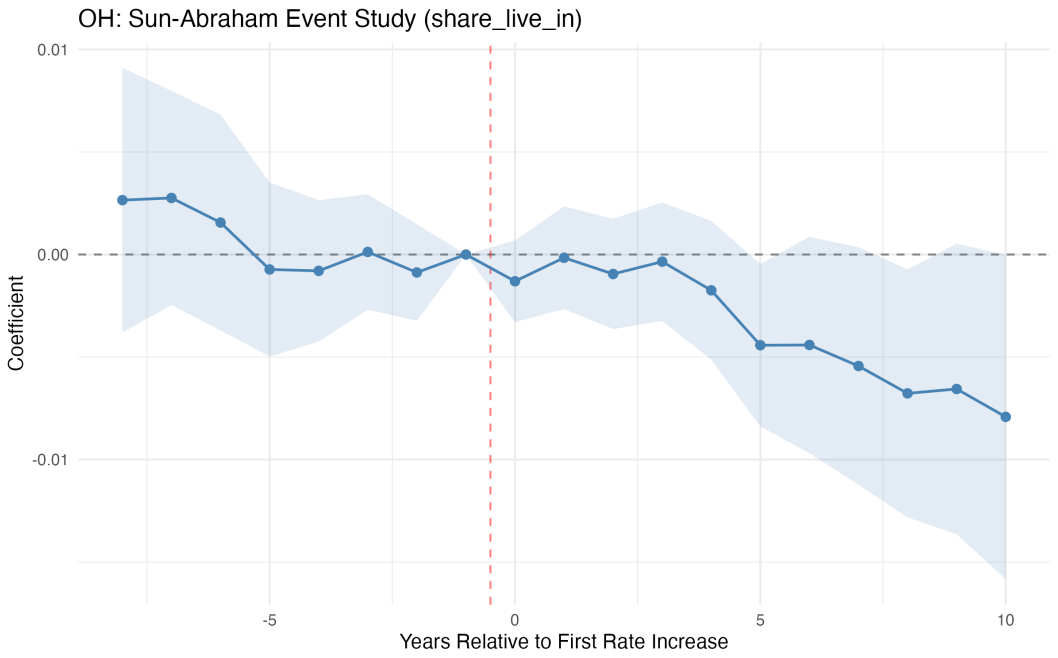


Figure 6: Ohio: Sun-Abraham Event Study (share_live_in)

5.2 Companion Outcomes: Decomposition

Table 3: Companion Outcomes: Decomposing the share_live_in Effect

	(1)	(2)	(3)	(4)
	share_live_in	log(Resident)	log(Nonresident)	log(Total Jobs)
Panel A: PA Dose-Response (treated only)				
rate × post				
post				

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

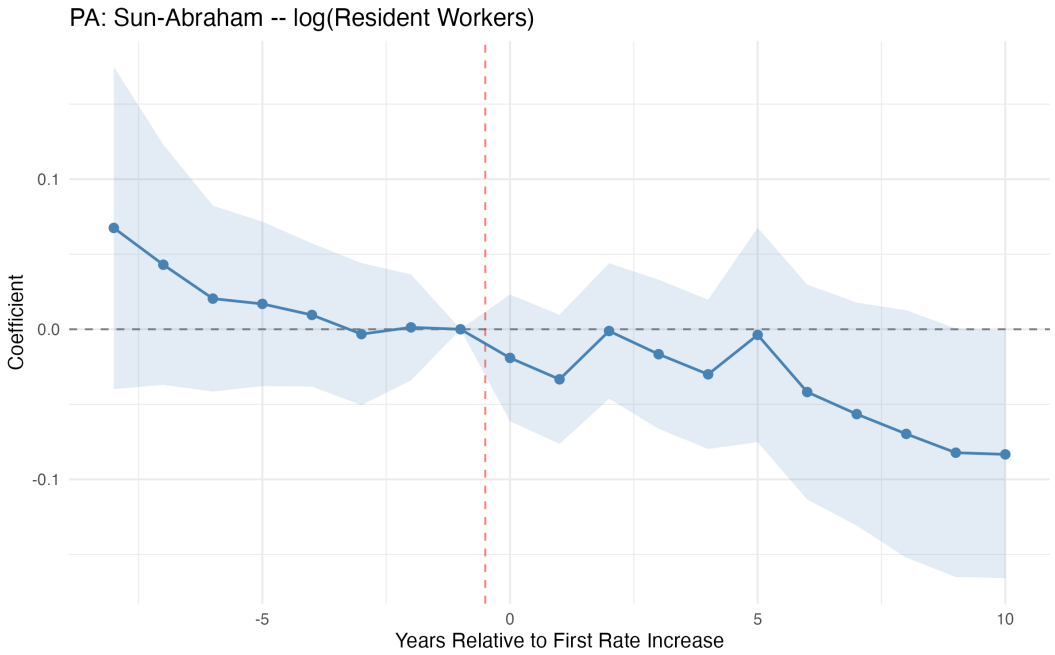


Figure 7: Pennsylvania: Sun-Abraham — log(Resident Workers)

5.3 Fiscal Impact

Table 4: Fiscal Impact of EIT Rate Changes

	(1)	(2)	(3)	(4)	(5)	(6)
	EIT Rev	Total Tax	Total Rev	Police	Population	Mkt Value
resident_eit						

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.4 Heterogeneity

6 Robustness

Table 5: Robustness: Fiscal Controls and Sample Restrictions

	(1)	(2)	(3)	(4)
	Baseline	+ Controls	No Philly	No Philly + Ctrl
resident_eit				

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7 Discussion

8 Conclusion

Acknowledgments

References

Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 6: Pre-Treatment Balance: Treated vs. Control Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	T N	C N
Population	14,174	3,711	0.139	1,265	39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267	39,692
EIT Revenue ($)	4.4M	438K	0.062		
Police Exp ($)	3.1M	424K	0.110		
EIT Rate (%)	0.5	0.5	0.055		

Notes:

B Additional Figures